KHALED DIAA ELDIN MOHAMED

Mechatronics Engineering Student

6 of October, Egypt | +201144775725 | khaliedworks@gmail.com | **LinkedIn** | **GitHub**

PROFILE

High-performing Mechatronics Engineering student (GPA **3.80**) with hands-on experience in **electromechanical design, automation systems, embedded communication, CFD/FEA analysis, and renewable energy systems**. Proven ability to integrate **mechanical, electrical, and control concepts** into functional engineering projects. Actively building an industry-oriented portfolio aligned with **industrial automation, robotics, and smart systems**.

EDUCATION

B.Sc. Mechatronics Engineering
Faculty of Engineering at MSA University — *Expected Graduation: 2028*

- **Cumulative GPA:** 3.80 / 4.00
- **Latest Semester GPA:** 3.86

ENGINEERING PROJECTS

1. Radar-Guided Navigation System for Autonomous Vehicles

- Designed an autonomous navigation system using a **stationary radar** for obstacle detection instead of onboard sensors
- Implemented **ESP32-based wireless communication (ESP-NOW)** for low-latency command transmission
- Developed real-time directional logic (Left / Right / Stay) with controlled response delay for stability
- Improved reliability under adverse conditions compared to camera or ultrasonic-based systems
  Technologies: ESP32, Radar Sensing, Wireless Communication, Embedded Systems

2. Centralized Grid-Connected Solar PV System

- Conducted **steady-state thermal simulation** of a rooftop PV module using **ANSYS Workbench** to evaluate temperature distribution and heat losses
- Linked thermal simulation results with **thermodynamic energy balance equations** to assess module efficiency under realistic operating conditions
- Estimated **annual energy yield** based on thermal performance and local solar conditions
- Performed **techno-economic analysis** including cost modeling and **ROI calculation** for residential rooftop PV systems under **Egyptian climate and market conditions**
- Integrated simulation outputs with **Excel-based engineering calculations** and validated results using material property data
    - **Tools & Methods:** ANSYS Workbench (Steady-State Thermal), Thermodynamics, Heat Transfer, Energy Balance, Excel Engineering Models

3. Six-Legged Spider Robot

- Designed and analyzed a **hexapod robotic platform** with coordinated leg motion
- Applied kinematics concepts to achieve stable gait and movement control
- Focused on mechanical robustness and balance for uneven motion scenarios

4. Three-Stage Gearbox Design

- Designed a multi-stage gearbox with torque and speed optimization
- Performed mechanical calculations for gear ratios, stresses, and efficiency
- Delivered complete CAD assemblies and engineering documentation
  Tools: SolidWorks, Mechanical Design Principles

5. IoT-Based Smart Irrigation System

- Developed a smart irrigation solution using environmental sensors and **IoT communication.**
- Automated irrigation decisions based on real-time data to improve water efficiency.
- Integrated sensing, control logic, and data transmission.

6. ANSYS CFD Analysis of Francis Turbine

- Performed computational fluid dynamics (CFD) analysis on a Francis turbine
- Analyzed velocity distribution, pressure contours, and flow efficiency
- Interpreted simulation results to assess turbine performance

7. ANSYS Structural Stress Analysis (Table Design)

- Conducted finite element analysis (FEA) on multiple table design configurations
- Compared stress distribution and deformation under varying load cases
- Optimized design for safety and structural efficiency

TECHNICAL SKILLS

- **Mechanical:** Mechanical Design & Machine Elements, Kinematics & Dynamics, SolidWorks.
- **Electrical & Control:** Electric Machines Fundamentals, Sensors & Measurements, Microcontrollers (ESP32).
- **Simulation:** ANSYS (CFD, Structural Analysis), MATLAB, Numerical Analysis.
- **Digital:** Wireless Communication (ESP-NOW), IoT Systems, Python, GitHub (Version Control).

CERTIFICATIONS & TRAINING

- **Automotive Diploma (Mechanical & Electrical Systems) (3 Months) — Enginest**
- **SolidWorks Training (30 Hours) — Engovation**
  (Certified & endorsed by the Egyptian Engineers Syndicate)

COMPETITIONS & ACTIVITIES

- **AAKRUTI 2025 – SolidWorks Competition**
  - Participated in a national-level mechanical design competition
  - Gained hands-on experience in design under constraints and evaluation standards

---

LANGUAGES

- **Arabic:** Native speaker
- **English:** B2
- **German:** A1